

gymbro

Setup Manual

A step-by-step guide to running the React Native workout tracker
on your own device.

Repository:

<https://github.com/gagangeet2517-arch/gymbro>

Built with Expo, React Native, and Expo Router

1. What is gymbro?

gymbro is a workout-tracking mobile app written with React Native, Expo, and Expo Router. It runs on iOS, Android, and the web from a single codebase. All workout data is stored locally on-device using AsyncStorage — there is no backend, no account system, and no internet connectivity required to use the app.

This manual walks through every step required to clone the project, install its dependencies, and run it on a simulator, emulator, or physical device.

2. Prerequisites

Before you start, make sure the following are installed on your machine.

2.1 Node.js (v20.19 or newer)

Expo SDK 54 requires Node 20.19+ or Node 22+. Verify your version:

```
node --version
```

If it is missing or too old, download the LTS installer from nodejs.org, or use a version manager such as `nvm` or `fnm`:

```
# Using nvm
nvm install 22
nvm use 22
```

2.2 Git

Required for cloning the repository.

```
git --version
```

On macOS, install via Xcode Command Line Tools:

```
xcode-select --install
```

On Windows, download from git-scm.com. On Linux, use your package manager (e.g. `sudo apt install git`).

2.3 A package manager

npm ships with Node, so nothing extra to install. If you prefer yarn or pnpm, they will work too — the examples in this guide use npm.

2.4 Platform tooling (pick at least one)

You need a way to actually see the app running. Pick whichever option fits your operating system and target platform:

- iOS Simulator (macOS only) — install Xcode from the Mac App Store, then open it once to accept the license. After that run ``xcode-select --install`` if you have not already.
- Android Emulator (any OS) — install Android Studio, open it, and use the AVD Manager to create a virtual device (Pixel 7, API 34 is a safe default). Make sure `ANDROID_HOME` points to your SDK folder.
- Physical phone with Expo Go — the fastest way to test on a real device. Install the free "Expo Go" app from the App Store or Play Store and scan the QR code Expo prints when you start the dev server. Your phone and computer must be on the same Wi-Fi network.
- Web browser — no extra tools needed; the app also runs as a web app for quick prototyping, though some native features will fall back gracefully.

Note: On macOS the recommended combo is the iOS Simulator (built in) plus Expo Go on your phone. On Windows or Linux, use the Android Emulator and/or Expo Go.

3. Cloning the repository

Open a terminal in the folder where you want the project to live, then run:

```
git clone https://github.com/gagangeet2517-arch/gymbro.git
cd gymbro
```

If you already have a local copy and just want to pull the latest changes:

```
cd gymbro
git pull origin main
```

4. Installing dependencies

From the project root, install everything listed in package.json:

```
npm install
```

This will pull down React, React Native, Expo, the navigation libraries, AsyncStorage, victory-native (charts), and the rest of the dependencies. The first install can take a few minutes.

Note: If the install fails with a peer-dependency error, try ``npm install --legacy-peer-deps``. If it fails with a permissions error, never use `sudo` — fix the npm prefix instead (see the troubleshooting section).

5. Running the app

All run commands start with ``npx expo`` so you do not need to install the Expo CLI globally.

5.1 Start the dev server

```
npx expo start
```

This launches the Metro bundler and shows a menu in your terminal along with a QR code. From here you can press a key to open the app in a specific environment:

- Press `i` to open the iOS Simulator (macOS).
- Press `a` to open the Android Emulator (or a connected device).
- Press `w` to open the web build in your default browser.
- Or scan the QR code with your phone (Camera app on iOS, the Expo Go app on Android) to load the app on a real device.

5.2 Open a specific platform directly

```
npx expo start --ios  
npx expo start --android  
npx expo start --web
```

5.3 Reload after code changes

Saving a file triggers Fast Refresh automatically. If state gets stuck, press `r` in the dev-server terminal to force a full reload, or shake the device to open the developer menu.

6. Verifying your setup

Once the dev server is running and the app loads, you should see the Home tab. To make sure everything is wired correctly:

- Navigate between the four bottom tabs: Home, Workouts, History, Progress. None of them should crash.
- Tap into Workouts — starter templates are seeded automatically on first launch.
- Start a workout, log a set, and finish it. Open the History tab to confirm the workout is saved.
- Force-quit and reopen the app. Your data should persist because it lives in AsyncStorage on the device.

7. Project structure (high level)

Knowing where things live helps when you start making changes:

- `app/` — file-based routes (Expo Router). `_layout.tsx` wraps the tree in three context providers. `(tabs)/` holds the bottom-tab screens.
- `context/` — `ExerciseContext`, `TemplateContext`, `WorkoutContext`. Each persists to AsyncStorage with a `hasHydrated` guard.
- `data/exerciseCatalog.ts` — the static catalog of built-in exercises and starter templates.
- `components/ui/` — shared primitives like `AppButton` and `AppCard`.
- `assets/` — icons, splash screens, fonts.

8. Useful scripts

Defined in `package.json`:

```
npm run start      # alias for `expo start`
npm run ios        # `expo run:ios` (full native build)
npm run android    # `expo run:android`
npm run web        # `expo start --web`
npm run lint       # ESLint
```

Type-check the project without compiling:

```
npx tsc --noEmit
```

9. Troubleshooting

9.1 Metro is serving stale code

Clear the bundler cache:

```
npx expo start -c
```

9.2 Weird native errors after pulling new code

Reinstall dependencies and pods (the iOS folder is checked in):

```
rm -rf node_modules
npm install
cd ios && pod install && cd ..
```

9.3 iOS Simulator never opens

Open Xcode once, accept the license, and make sure the command-line tools point at the full Xcode (not just the standalone tools):

```
sudo xcode-select -s /Applications/Xcode.app/Contents/Developer
```

9.4 Android emulator says "unable to connect"

Make sure an AVD is actually running (Android Studio > Device Manager). If the device list is empty:

```
adb devices
```

If adb is not on your PATH, add the Android SDK platform-tools folder to your shell profile (e.g. ~/.zshrc):

```
export PATH=$PATH:$HOME/Library/Android/sdk/platform-tools
```

9.5 QR code scan fails on a real phone

- Confirm the phone and computer are on the same Wi-Fi.
- Some corporate or public networks block the required ports; try a personal hotspot or use tunnel mode: `npx expo start --tunnel`.
- If Expo Go cannot load a JS bundle, fully close the Expo Go app and rescan.

9.6 npm permission errors during install

Do not run npm with sudo. Instead, point the npm prefix at a folder you own:

```
mkdir -p ~/.npm-global
npm config set prefix '~/.npm-global'
export PATH=~/.npm-global/bin:$PATH    # add to ~/.zshrc
```

10. Pulling in updates

To grab the latest changes from GitHub:

```
git pull origin main
npm install          # in case new dependencies were added
```

If the Expo SDK version changes between pulls, also run:

```
npx expo install --check
```

This audits installed native modules against the SDK version and tells you which ones need to be updated.

11. Pushing your own changes

Make a feature branch, commit, and push:

```
git checkout -b my-feature
# ... make changes ...
git add .
git commit -m "Describe what you changed"
git push -u origin my-feature
```

Then open a pull request on GitHub to merge your branch into main.

12. Where to get help

- Expo documentation — docs.expo.dev
- React Native documentation — reactnative.dev
- Expo Router — docs.expo.dev/router/introduction
- Open an issue at github.com/gagangeet2517-arch/gymbro/issues for anything specific to this project.

End of manual. If you spot something out of date in this document, edit `SETUP.pdf` (or the generator that produced it) and open a pull request.